

TAYLOR'S THEOREM

Lesson 19

Theorem:

If $f(x)$ and its n derivatives $f', f'', \dots, f^{(n)}$ are continuous on $[a, b]$ and $f^{(n)}$ is differentiable on (a, b) then there exists $c \in (a, b)$ such that:

$$f(b) = f(a) + \frac{f'(a)}{1!}(b-a) + \frac{f''(a)}{2!}(b-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(b-a)^n + \frac{f^{(n+1)}(c)}{(n+1)!}(b-a)^{n+1}$$

NB: Taylor's Theorem is a generalization of the Mean Value theorem.

TAYLOR SERIES

If $f(x)$ has derivatives of all orders in an open interval I , containing a , then for each positive integer n and for each x in I ,

$$f(x) = f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \frac{1}{3!}f'''(a)(x-a)^3 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

$$\text{Where } R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \text{ for } a < c < x$$

Note:

$$f(x) = f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \frac{1}{3!}f'''(a)(x-a)^3 + \dots + \frac{1}{n!}f^{(n)}(a)(x-a)^n$$

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

Remainder of order n .

Taylor series of degree n at $x=a$.

As $n \rightarrow \infty$, we have

$$* f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)(x-a)^k}{k!} \leftarrow \begin{cases} \text{Taylor series of} \\ f(x) \text{ at } x=a \end{cases}$$

$$* f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)(x-a)^k}{k!} \leftarrow \begin{cases} \text{Taylor polynomial of degree } n \\ \text{at } x=a \end{cases}$$

The remainder of Taylor's Series $R_n(x)$ is estimated using the theorem below:

THEOREM: (Remainder Estimation Theorem)

If there is a positive constant M such that $|f^{(n+1)}(t)| \leq M$ for all t between x and a , inclusive, then the remainder term $R_n(x)$ in Taylor's Theorem satisfies the inequality

$$|R_n(x)| \leq M \frac{|x-a|^{n+1}}{(n+1)!}$$

if this inequality holds for every n and the other conditions of Taylor's Theorem are satisfied by f , then the series converges to $f(x)$.

* NOTE: Taylor series of $f(x)$ at $x=0$ is ^{called} ~~the~~ Maclaurin series of $f(x)$.

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)(x-a)^k}{k!}$$

"Taylor series of $f(x)$ at $x=a$."

Put $x=0$

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)x^k}{k!}$$

"Maclaurin series of $f(x)$ "

"ie Taylor series of $f(x)$ at $x=0$."

Example 1

Find the Taylor series of $\ln x$ at $x=1$ and hence deduce the Taylor series of $\frac{\ln x}{x-1}$.

Solution

Let $f(x) = \ln x$

at $x=1$

$$f(1) = \ln 1 = 0$$

$$f'(x) = \frac{1}{x}$$

$$f'(1) = \frac{1}{1} = 1$$

$$f''(x) = -\frac{1}{x^2}$$

$$f''(1) = -\frac{1}{1^2} = -1$$

$$f'''(x) = \frac{2}{x^3}$$

$$f'''(1) = \frac{2}{1} = 2$$

$$f^{(4)}(x) = -\frac{6}{x^4}$$

$$f^{(4)}(1) = -\frac{6}{1} = -6$$

Taylor series of $f(x)$ at $x=1$.

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$$f(x) = f(1) + f'(1)(x-1) + \frac{1}{2!} f''(1)(x-1)^2 + \frac{1}{3!} f'''(1)(x-1)^3 + \frac{1}{4!} f^{(4)}(1)(x-1)^4 + \dots$$

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{1}{3!} (2)(x-1)^3 + \frac{1}{4!} (-6)(x-1)^4 + \dots$$

$$\therefore \ln x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \dots$$

Dividing by $(x-1)$ both sides, we have:

$$\frac{\ln x}{x-1} = 1 - \frac{1}{2}(x-1) + \frac{1}{3}(x-1)^2 - \frac{1}{4}(x-1)^3 + \dots$$

Example 2

Find the Maclaurin series of the following functions

(a) $f(x) = \sin x$

(b) $g(x) = \ln(1+x)$

Solution:

At $x=0$

(a) $f(x) = \sin x$, $f(0) = 0$
 $f'(x) = \cos x$, $f'(0) = 1$
 $f''(x) = -\sin x$, $f''(0) = 0$
 $f'''(x) = -\cos x$, $f'''(0) = -1$
 $f^{(4)}(x) = \sin x$, $f^{(4)}(0) = 0$

Maclaurin of $f(x)$ is

$$f(x) = f(0) + f'(0)x + \frac{1}{2!} f''(0)x^2 + \frac{1}{3!} f'''(0)x^3 + \dots$$

$$\sin x = x - \frac{1}{3!} x^3 + \frac{1}{5!} x^5 - \frac{1}{7!} x^7 + \dots$$

$$\sin x = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^{2k-1}}{(2k-1)!}$$

(b) $g(x) = \ln(1+x)$, $g(0) = \ln 1 = 0$

$$g'(x) = \frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots, g'(0) = 1$$

$$g''(x) = -1 + 2x - 3x^2 + 4x^3 - \dots, g''(0) = -1$$

$$g'''(x) = 2 - 6x + 12x^2 - \dots, g'''(0) = 2$$

$$g^{(4)}(x) = -6 + 24x - \dots, g^{(4)}(0) = -6$$

Maclaurin of $g(x)$ is

$$g(x) = g(0) + g'(0)x + \frac{1}{2!} g''(0)x^2 + \frac{1}{3!} g'''(0)x^3 + \frac{1}{4!} g^{(4)}(0)x^4 + \dots$$

$$\ln(1+x) = x - \frac{1}{2} x^2 + \frac{1}{3} x^3 - \frac{1}{4} x^4 + \dots$$

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1} x^k}{k}$$

First 4 non-zero terms in ascending order of x
 Complete series to ∞ .

MACLAURIN SERIES OF COMMON FUNCTIONS

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$$(1) \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + x^5 + x^6 + \dots$$

$$(2) \frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + x^6 - \dots$$

$$(3) \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \frac{x^{11}}{11!} + \dots$$

$$(4) \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \frac{x^{10}}{10!} + \dots$$

$$(5) \tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

$$(6) \sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \frac{x^9}{9!} + \dots$$

$$(7) \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \dots$$

$$(8) \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \frac{x^6}{6} + \dots$$

$$(9) e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

Example 3.

Find the first 4 terms of $e^x \sinh x$ in ascending power of x .

Solution

$$\begin{aligned} * \text{ We have: } e^x \sinh x &= \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots\right) \left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots\right) \\ &= x + x^2 + \frac{x^3}{2} + \frac{x^4}{6} + \dots \\ &\quad + \frac{x^3}{6} + \frac{x^4}{6} + \dots \end{aligned}$$

$$= x + x^2 + \left(\frac{1}{2} + \frac{1}{6}\right)x^3 + \left(\frac{1}{6} + \frac{1}{6}\right)x^4 + \dots$$

$$\therefore e^x \sinh x = x + x^2 + \frac{2}{3}x^3 + \frac{1}{3}x^4 + \dots$$

Example 4(a) Use Maclaurin series of $\tan x$ to compute the limit

$$\lim_{x \rightarrow 0} \left\{ \frac{\tan x - x}{x^3} \right\}$$

(b) Use L'Hopital method.

Solution(a) Maclaurin Series of $\tan x$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

$$\text{ie } \tan x - x = \frac{x^3}{3} + \frac{2x^5}{15} + \dots$$

$$\frac{\tan x - x}{x^3} = \frac{1}{3} + \frac{2x^2}{15} + \dots$$

$$\lim_{x \rightarrow 0} \left\{ \frac{\tan x - x}{x^3} \right\} = \frac{1}{3}$$

$$(b) \lim_{x \rightarrow 0} \left\{ \frac{\tan x - x}{x^3} \right\} = \frac{0}{0} \quad \text{indeterminate form}$$

Apply L'Hopital

$$\lim_{x \rightarrow 0} \left\{ \frac{\sec^2 x - 1}{3x^2} \right\} = \frac{0}{0}$$

Apply L'Hopital

$$\lim_{x \rightarrow 0} \left\{ \frac{2 \sec x (\sec x \tan x)}{6x} \right\} = \frac{0}{0}$$

Apply L'Hopital again

$$\lim_{x \rightarrow 0} \left\{ \frac{4 \sec^2 x \tan x + 2 \sec^4 x}{6} \right\} = \frac{2}{6} = \frac{1}{3}$$

$$\therefore \lim_{x \rightarrow 0} \left\{ \frac{\tan x - x}{x^3} \right\} = \frac{1}{3}$$

Example 5Use Maclaurin series of $\sinh x$ to compute the limit

$$\lim_{x \rightarrow 0} \left\{ \frac{\sinh x - x}{x^3} \right\}$$

Solution

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \frac{x^9}{9!} + \dots$$

$$\sinh x - x = \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots$$

$$\frac{\sinh x - x}{x^3} = \frac{1}{3!} + \frac{x^2}{5!} + \frac{x^4}{7!} + \dots$$

$$\therefore \lim_{x \rightarrow 0} \left\{ \frac{\sinh x - x}{x^3} \right\} = \frac{1}{6}$$